

EX. NO: 7

IMPLEMENTATION OF RSA ALGORITHM

AIM

To write a C program to implement the RSA (Rivest–Shamir–Adleman) encryption algorithm.

ALGORITHM

Step 1: Start the program.

Step 2: Select two prime numbers p and q.

Step 3: Compute:

- $n = p \times q$
- $\phi(n) = (p - 1)(q - 1)$

Step 4: Choose a public key e such that:

- $1 < e < \phi(n)$
- e and $\phi(n)$ are coprime.

Step 5: Calculate the private key d such that:

$$d \times e \equiv 1 \pmod{\phi(n)}$$

Step 6: Read the plaintext message from the user.

Step 7: Encrypt the plaintext using:

$$C = M^e \pmod{n}$$

Step 8: Decrypt the ciphertext using:

$$M = C^d \pmod{n}$$

Step 9: Display the encrypted and decrypted messages.

Step 10: Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
long int power(long int a, long int b, long int mod)
```

```
{
```

```
    long int result = 1;
```

```
    while(b > 0)
```

```
    {
```

```
        result = (result * a) % mod;
```

```
        b--;
```

```

    }
    return result;
}

int main()
{
    long int p, q, n, phi, e, d, msg, cipher, decrypt;
    printf("Enter two prime numbers (p and q): ");
    scanf("%ld %ld", &p, &q);
    n = p * q;
    phi = (p - 1) * (q - 1);
    printf("Enter public key (e): ");
    scanf("%ld", &e);
    // Find private key d
    d = 1;
    while((d * e) % phi != 1)
        d++;
    printf("Enter message (number less than %ld): ", n);
    scanf("%ld", &msg);
    cipher = power(msg, e, n);
    decrypt = power(cipher, d, n);
    printf("\nPublic Key (e,n): (%ld,%ld)", e, n);
    printf("\nPrivate Key (d,n): (%ld,%ld)", d, n);
    printf("\nEncrypted Message : %ld", cipher);
    printf("\nDecrypted Message : %ld\n", decrypt);
    return 0;
} OUTPUT:

```

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main.c

10 scanf("%s", plaintext);

11

12 printf("Enter Key: ");

13 scanf("%s", key);

14

15 int keyLen = strlen(key);

16

17 // Simple XOR-based demonstration

18 for(i = 0; plaintext[i] != '\0'; i++)

19 {

20 cipher[i] = plaintext[i] ^ key[i % keyLen];

21 }

22

23 cipher[i] = '\0';

24

25 printf("\nEncrypted Cipher (Hex): ");

26

27 for(i = 0; i < strlen(plaintext); i++)

28 printf("%02X ", (unsigned char)cipher[i]);

29

30 printf("\n");

31

32 return 0;

33 }

Run

Share

Output

Clear

Enter Plain Text: pallavi

Enter Key: key

Encrypted Cipher (Hex): 18 04 15 07 04 0F 02

=== Code Execution Successful ===

27°C Mostly cloudy

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